

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE274	APPLIED MACHINE LEARNING	2	0	2	3	
Course Weightage	ATT: 5 CA: 25 MTT: 20 ETT: 50					

Course Outcomes :Through this course students should be able to

CO1 :: explain the types of data and data pre-processing concepts

CO2 :: apply feature engineering and dimensionality reduction techniques for machine learning models.

CO3 :: compare and Implement nonlinear and distance-based models using appropriate ROC- and precision-recall–based evaluation metrics.

CO4 :: apply linear, regularized, and tree-based regression models using standard regression performance metrics.

CO5 :: apply ensemble learning models through pipelines and systematic hyperparameter tuning techniques.

CO6 :: perform clustering using K-means, hierarchical, and density-based methods and evaluate results using established clustering metrics.

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	MACHINE LEARNING FOR ABSOLUTE BEGINNERS: A PLAIN ENGLISH INTRODUCTION (THIRD EDITION)	OLIVER THEOBALD	NULL

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	MACHINE LEARNING	S SRIDHAR, M VIJAYALAKSHMI	OXFORD UNIVERSITY PRESS

Other Reading (OR)	
Sr No	Journals articles as Compulsary reading (specific articles, complete reference)
OR-1	https://www.pvpsiddhartha.ac.in/dep_it/lecture%20notes/3-2-23/MLT/ML%20UNIT%202.pdf ,
OR-2	https://drgupopeengg.org/wp-content/uploads/2023/09/Unit-3-AIML.pdf ,

Relevant Websites (RW)		
Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	https://www.iitr.ac.in/media/facspace/patelfec/16Bit/slides/Lecture-2-Data-Preprocessing-Part-1.pdf	Data Preprocessing

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RW-2	https://www.geeksforgeeks.org/machine-learning/what-is-feature-engineering/	Feature engineering and dimensionality reduction
RW-3	https://www.ibm.com/think/topics/classification-vs-regression	Regression
RW-4	https://pingax.com/projects/data-scientist/machine-learning/classification-vs-regression-models-a-practical-example/	Regression
RW-5	https://www.geeksforgeeks.org/machine-learning/a-comprehensive-guide-to-ensemble-learning/	Ensemble learning
RW-6	https://medium.com/@fraidoonomarzai99/hyperparameters-tunning-and-cross-validation-in-depth-d0918b62d986	Hyperparameter tuning

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	4

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Data Pre-processing(Types of data, Data pre-processing, data leakage concept)	T-1	RW-1	Lecture 1 will be considered as Lecture Zero	Students will learn about Data Pre-processing	PPT	
	Lecture 2	Data Pre-processing (handling missing values, outliers handling,)	T-1 R-1	RW-1	Lecture 2 will be used to discuss about Types of data, Data pre-processing, data leakage concept,handling missing values	Students will learn about data pre-processing	PPT	
Week 2	Lecture 3	Data Pre-processing (handling categorical data, scaling and normalization)	T-1 R-1	RW-1	Lecture 3 will be used to discuss handling categorical data, scaling and normalization	Students will learn about Data preprocessing	PPT, Live examples	
	Lecture 4	Data Pre-processing(class imbalance handling)	T-1 R-1		Lecture 4 will be used to discuss class imbalance handling	Students will learn about Data preprocessing	PPT, Live examples	

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Week 3	Lecture 6	Feature Engineering(tree-based feature importance)	T-1 R-1	RW-2	Lecture 6 will be used to discuss Feature Extraction, creating new features, aggregation features	Students will learn about Feature Engineering	PPT,Live Examples	
		Feature Engineering(Feature Extraction, creating new features, aggregation features)	T-1 R-1	RW-2	Lecture 6 will be used to discuss Feature Extraction, creating new features, aggregation features	Students will learn about Feature Engineering	PPT,Live Examples	
Week 4	Lecture 7				Test - Code based			
	Lecture 8	Dimensionality Reduction (Curse of dimensionality,)	T-1	RW-2	Lecture 8 will be used to discuss about Principal Component Analysis	Students will learn about Dimensionality Reduction	PPT, Live examples	
		Dimensionality Reduction (Principal Component Analysis)	T-1	RW-2	Lecture 8 will be used to discuss about Principal Component Analysis	Students will learn about Dimensionality Reduction	PPT, Live examples	
Week 5	Lecture 9	Dimensionality Reduction (explained variance ration, Linear Discriminant Analysis)	T-1	RW-2	Lecture 9 will be used to discuss about explained variance ration, Linear Discriminant Analysis	Students will learn about Dimensionality Reduction	PPT, Live examples	
	Lecture 10	Linear & Probabilistic Classification Models (Perceptron, Logistic Regression)	T-1	OR-1 OR-2	Lecture 10 will be used to discuss about Perceptron, Logistic Regression, Naïve Bayes Classifier	Students will learn about Linear & Probabilistic Classification Models	PPT, Live examples	
		Linear & Probabilistic Classification Models (Introduction to supervised Learning)	T-1	OR-1 OR-2	Lecture 10 will be used to discuss about Introduction to supervised Learning Perceptron, Logistic Regression	Students will learn about Linear & Probabilistic Classification Models	PPT, Live examples	
Week 6	Lecture 11	Linear & Probabilistic Classification Models(Naïve Bayes Classifier)	T-1	OR-1 OR-2	Lecture 3 will be used to discuss Naïve Bayes Classifier,Model evaluation using confusion matrix, precision, recall, F1-Score	Students will learn about Linear & Probabilistic Classification Models	PPT, Live examples	
		Linear & Probabilistic Classification Models (Model evaluation using confusion matrix, precision, recall, F1-Score)	T-1	OR-1 OR-2	Lecture 3 will be used to discuss Naïve Bayes Classifier,Model evaluation using confusion matrix, precision, recall, F1-Score	Students will learn about Linear & Probabilistic Classification Models	PPT, Live examples	

Week 6	Lecture 12	Nonlinear & Distance-Based Models(K-Nearest Neighbors, Decision Tree)	T-1	OR-1 OR-2	Lecture 12 will be used to discuss K-Nearest Neighbors, Decision Tree	Students will learn about Nonlinear & Distance-Based Models	PPT, Live examples	
Week 7	Lecture 13	Nonlinear & Distance-Based Models(Support Vector Machine, Model evaluation using ROC Curve)	T-1	OR-1 OR-2	Lecture 2 will be used to discuss Support Vector Machine, Model evaluation using ROC Curve ,ROC-AUC, Precision-Recall Curve, AUC-PR	Students will learn about Nonlinear & Distance-Based Models	PPT, Live	
		Nonlinear & Distance-Based Models(ROC-AUC, Precision-Recall Curve, AUC-PR)	T-1	OR-1 OR-2	Lecture 2 will be used to discuss Support Vector Machine, Model evaluation using ROC Curve ,ROC-AUC, Precision-Recall Curve, AUC-PR	Students will learn about Nonlinear & Distance-Based Models	PPT, Live	
		SPILL OVER						
Week 7	Lecture 14				Spill Over			
		MID-TERM						
Week 8	Lecture 15	Regression(Difference between regression and classification, Bias-variance considerations in regression)	T-1	RW-3 RW-4	Lecture 15 will be used to discuss about Difference between regression and classification, Bias-variance considerations in regression ,Simple Linear Regression, Multiple Linear Regression	Students will learn about Regression	PPT, Live examples	
		Regression(Simple Linear Regression, Multiple Linear Regression)	T-1	RW-3 RW-4	Lecture 15 will be used to discuss about Difference between regression and classification, Bias-variance considerations in regression ,Simple Linear Regression, Multiple Linear Regression	Students will learn about Regression	PPT, Live examples	

Week 8	Lecture 16	Regression(Interpretation of coefficients, Regularized Regression models,)	T-1	RW-3 RW-4	Lecture 16 will be used to discuss about Interpretation of coefficients, Regularized Regression models,	Students will learn about Regression	PPT, Live examples	
Week 9	Lecture 17	Regression(Effect of regularization on model complexity)	T-1	RW-3 RW-4	Lecture 17 will be used to discuss about Effect of regularization on model complexity ,Polynomial feature expansion, Tree-Based regression models	Students will learn about Regression	PPT, Live examples	
		Regression(Polynomial feature expansion, Tree-Based regression models)	T-1	RW-3 RW-4	Lecture 17 will be used to discuss about Effect of regularization on model complexity ,Polynomial feature expansion, Tree-Based regression models	Students will learn about Regression	PPT, Live examples	
	Lecture 18	Regression(Time-series Regression models)	T-1	RW-3 RW-4	Lecture 18 will be used to discuss about Time-series Regression models	Students will learn about Regression	PPT, Live examples	
Week 10	Lecture 19	Ensemble Learning (Introduction to ensemble learning)	T-1	RW-5	Lecture 19 will be used to discuss Introduction to ensemble learning, Bagging & Boosting Ensembles, majority voting classifier	Students will learn about Ensemble Learning	PPT, Live examples	
		Ensemble Learning(Bagging & Boosting Ensembles, majority voting classifier)	T-1	RW-5	Lecture 19 will be used to discuss Introduction to ensemble learning, Bagging & Boosting Ensembles, majority voting classifier	Students will learn about Ensemble Learning	PPT, Live examples	
	Lecture 20	Ensemble Learning(Random Forest, AdaBoost, Gradient Boosting Machines)	T-1	RW-5	Lecture 20 will be used to discuss about Random Forest, AdaBoost, Gradient Boosting Machines XGBoost, LightGBM, CatBoost, Ensemble Regression Models	Students will learn about Ensemble Learning	PPT, Live examples	

Week 10	Lecture 20	Ensemble Learning (XGBoost, LightGBM, CatBoost, Ensemble Regression Models)	T-1	RW-5	Lecture 20 will be used to discuss about Random Forest, AdaBoost, Gradient Boosting Machines XGBoost, LightGBM, CatBoost, Ensemble Regression Models	Students will learn about Ensemble Learning	PPT, Live examples	
Week 11	Lecture 21	Model evaluation and hyperparameter tuning (Pipelines, Cross-validation strategies)	T-1	RW-6	Lecture 2 will be used to discuss about Pipelines, Cross-validation strategies	Students will learn about Model evaluation and hyperparameter tuning	PPT, Live examples	
	Lecture 22	Model evaluation and hyperparameter tuning(Grid Search, Random Search, Bayesian Optimization)	T-1		Lecture 22 will be used to discuss about Grid Search, Random Search, Bayesian Optimization	Students will learn about Model evaluation and hyperparameter tuning	PPT, Live examples	
Week 12	Lecture 23	Unsupervised Learning(Role of unsupervised learning, Differences between supervised and unsupervised learning)	T-1		Lecture 2 will be used to discuss about Role of unsupervised learning, Differences between supervised and unsupervised learning, Euclidean, Manhattan, Cosine distances	Students will learn about Unsupervised Learning	PPT, Live examples	
		Unsupervised Learning (Euclidean, Manhattan, Cosine distances)	T-1		Lecture 2 will be used to discuss about Role of unsupervised learning, Differences between supervised and unsupervised learning, Euclidean, Manhattan, Cosine distances	Students will learn about Unsupervised Learning	PPT, Live examples	
	Lecture 24	Unsupervised Learning (Choosing appropriate distance metrics)	T-1		Lecture 24 will be used to discuss Choosing appropriate distance metrics	Students will learn about Unsupervised Learning	PPT, Live examples	
Week 13	Lecture 25	Clustering Algorithms(K-Means Clustering, Elbow method, K-Medoids)	T-1		Lecture 25 will be used to discuss Density-Based Clustering, Hierarchical Clustering, linkage criteria, K-Means Clustering, Elbow method, K-Medoids	Students will learn about Clustering Algorithms	PPT, Live examples	

Week 13	Lecture 25	Clustering Algorithms (Hierarchical Clustering, linkage criteria)	T-1		Lecture 25 will be used to discuss Density-Based Clustering, Hierarchical Clustering, linkage criteria, K-Means Clustering, Elbow method, K-Medoids	Students will learn about Clustering Algorithms	PPT, Live examples	
		Clustering Algorithms (Density-Based Clustering)	T-1		Lecture 25 will be used to discuss Density-Based Clustering, Hierarchical Clustering, linkage criteria, K-Means Clustering, Elbow method, K-Medoids	Students will learn about Clustering Algorithms	PPT, Live examples	
	Lecture 26	Clustering Algorithms (Anomaly Detection)	T-1		Lecture 2 will be used to discuss about Silhouette Score, Davies–Bouldin Index, Evaluating clustering algorithms using Inertia, Anomaly Detection	Students will learn about Clustering Algorithms	PPT, Live examples	
		Clustering Algorithms (Evaluating clustering algorithms using Inertia)	T-1		Lecture 2 will be used to discuss about Silhouette Score, Davies–Bouldin Index, Evaluating clustering algorithms using Inertia, Anomaly Detection	Students will learn about Clustering Algorithms	PPT, Live examples	
		Clustering Algorithms (Silhouette Score, Davies–Bouldin Index)	T-1		Lecture 2 will be used to discuss about Silhouette Score, Davies–Bouldin Index, Evaluating clustering algorithms using Inertia, Anomaly Detection	Students will learn about Clustering Algorithms	PPT, Live examples	
Week 14	Lecture 27				Project			
		SPILL OVER						
Week 14	Lecture 28				Spill Over			
Week 15	Lecture 29				Spill Over			
	Lecture 30				Spill Over			

Scheme for CA:

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CA Category of this Course Code is:A0202 (2 out of 2)

Component	Weightage (%)	Mapped CO(s)
Recoup CA		
Test - Code based	33.34	CO1, CO2, CO3
Project	66.66	CO1, CO2, CO3, CO4, CO5, CO6

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Project	To check the student learnability of Applied ML.	To implement all the topics in apractical way	Group	Online	30	3 / 12
Test - Code based	To check the student learnability of Applied ML	Syllabus will be UNIT-1,2,and 3	Individual	Offline	30	5 / 6
Recoup CA	subjective test based on the syllabus	Additional chance of CA	Individual	Online	30	8 / 10

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	List of Practicals	Using a real-world dataset, perform data pre-processing by identifying data types, handling missing values, detecting and treating outliers, encoding categorical variables, and applying scaling and normalization techniques.		Student will learn to perform data pre-processing

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Practical 2	List of Practicals	Using a classification dataset, demonstrate handling of class imbalance using under-sampling, over-sampling, and SMOTE techniques, and illustrate data leakage scenarios with proper train–test splitting.		Student will Learn to implement classification problem
Practical 3	List of Practicals	Apply feature selection techniques including variance thresholding, correlation-based feature removal, forward selection, backward elimination, and tree-based feature importance on a given dataset.		Student will learn to Apply feature selection techniques
Practical 4	List of Practicals	Implement dimensionality reduction using PCA and LDA on a high-dimensional dataset, analyze explained variance ratio, and visualize data in reduced dimensions.		Student will learn to Implement dimensionality reduction
Practical 5	List of Practicals	Implement Perceptron, Logistic Regression, and Naïve Bayes classifiers on a labeled dataset, and evaluate model performance using confusion matrix, precision, recall, and F1-score.		Student will learn to Implement Perceptron, Logistic Regression, and Naïve Bayes classifiers
Practical 6	List of Practicals	Implement K-Nearest Neighbors, Decision Tree, and Support Vector Machine classifiers, and evaluate the models using ROC curve, ROC-AUC, Precision–Recall curve, and AUC-PR.		Student will learn to Implement K-Nearest Neighbors, Decision Tree, and Support Vector Machine classifiers,
Practical 7	List of Practicals	Implement regularized regression , polynomial regression, and tree-based regression models, and evaluate performance using MAE, MSE, RMSE, R ² score, residual plots, and actual vs predicted plots.		Student will learn to Implement regularized regression
Practical 8	List of Practicals	Implement ensemble classification models including majority voting, Random Forest, AdaBoost, Gradient Boosting, XGBoost, LightGBM, and CatBoost, and compare their performance with individual base learners.		Student will learn to Implement ensemble classification models
Practical 9	List of Practicals	Build machine learning pipelines and apply cross-validation, Grid Search, Random Search, and Bayesian Optimization to tune hyperparameters and improve model performance.		Student will learn to Build machine learning pipelines

Practical 10	List of Practicals	Implement clustering algorithms including K-Means, K-Medoids, and Hierarchical Clustering, determine the optimal number of clusters using the Elbow method, and analyze linkage criteria.		Student will learn to Implement clustering algorithms
	SPILL OVER			
Practical 11	Spill Over			